Civil Engineering Hydraulics and Geotechnical Reasoning

CIVIL-ENGINEERING · ce-hydraulics-geotech · BoardPrep lesson v1

Official examination scope

The current PRC CELE program lists fluid properties and flow, hydrostatics, water supply, soil classification, seepage, soil strength, bearing, compaction, settlement, earth pressure, and slope stability. This guide teaches core reasoning and selected calculations; project design requires site data, applicable standards, and qualified review.

Learning objectives

- Distinguish pressure, force, head, and flow using units.
- Explain effective stress and why drainage conditions matter.
- Estimate a footing area from the allowable pressure explicitly supplied in a problem.

1. Fluid pressure, head, and flow

For a static fluid of constant density, gauge pressure increases with depth according to p = gh; absolute pressure adds the local atmospheric pressure. Pressure is force per area, while hydraulic head expresses energy per unit weight as a length. For steady incompressible flow in a closed system, continuity relates discharge to area and average velocity, Q = AV. Bernoulli-type energy accounting also requires the correct elevation, pressure, velocity, pump, turbine, and loss terms. The simplified relations are useful only when their assumptions fit the flow; real channels and networks require appropriate resistance and boundary information.

2. Soil phases, water, and effective stress

Soil behavior depends on the relationships among solids, water, and air, not on a single label such as sand or clay. Water content, void ratio, degree of saturation, and unit weight describe different properties and must not be substituted for one another. For saturated soil under the usual soil-mechanics convention, effective stress ´ = - u, where is total stress and u is pore-water pressure. Changes in pore pressure can therefore change the stress carried by the soil skeleton even when the overlying load is unchanged. The convention and drainage conditions should be stated before interpreting strength or settlement.

3. Bearing, settlement, and slope stability

Bearing checks begin with a clearly defined load and contact area; q = P/A gives average contact pressure only for the assumed distribution. An allowable pressure already embeds assumptions supplied by the problem or governing geotechnical report, so it is not a universal soil constant. Settlement is a deformation problem whose magnitude and time depend on compressibility, drainage path, and stress change. Lateral earth pressure and slope stability likewise depend on geometry, water, soil strength, and construction sequence. Distinguish an ultimate failure calculation from a serviceability limit and do not infer site capacity from a classroom example.

4. Investigation, compaction, and interpretation

A soil description is a hypothesis supported by sampling, testing, and field observations. Classification organizes material using observed and measured properties; it does not replace a strength or compressibility test for a design decision. Compaction changes density by reducing air voids, and the useful moisture-density relationship is established through a stated test method. For seepage, identify the hydraulic boundary and flow path before applying a gradient or discharge model. Record test method, sample disturbance, groundwater conditions, and limitations so another engineer can see how the interpretation was reached.

Estimate a footing area

Question: A footing carries 900 kN on soil with an allowable bearing pressure of 150 kPa. What minimum plan area follows from the stated average-pressure model?
1. Use $q = P/A$ and rearrange $A = P/q$.
2. Remember 1 kPa = 1 kN/m², so $A = 900\ \text{kN} \div 150\ \text{kN/m}^2$.
3. Check that force divided by pressure leaves square metres.
Answer: 6 m² minimum area under the supplied allowable-pressure assumption.
Check: This does not determine shape, eccentricity, settlement, depth, or code compliance.

Common misconceptions

- Total stress and effective stress always change together by the same amount.
- A problem’s allowable pressure is universally valid for another site.
- A soil classification alone proves settlement or slope safety.

Practice patterns

- Write the assumed water condition before using a pressure relation.
- Keep total stress, pore pressure, and effective stress separate.
- Check the model limit before turning a calculated value into a design conclusion.

Official scope references

Official scope: Revised Program of the Civil Engineers Licensure Examination March 2026 — Professional Regulation Commission; Professional Regulatory Board of Civil Engineering — https://www.prc.gov.ph/sites/default/files/RevisedExamProgramMarch2026CivilEngg.pdf

Instructional references

Instructional reference: Hydraulic Design of Flood Control Channels — U.S. Army Corps of Engineers — https://www.publications.usace.army.mil/Portals/76/Publications/EngineerManuals/EM_1110-2-1601.pdf
Instructional reference: Manual of Hydrology: General Surface-Water Techniques — U.S. Geological Survey — https://pubs.usgs.gov/wsp/1541a/report.pdf
Instructional reference: Soils and Foundations Reference Manual — Federal Highway Administration, National Highway Institute — https://highways.dot.gov/sites/fhwa.dot.gov/files/FHWA-NHI-06-089.pdf

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